

WAHPETON, ND, USA

WMO: 722004

Lat: 46.244N

Lon: 96.607W

Elev: 295

StdP: 97.83

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.0	(c) -26.0	(d) -32.1	(e) 0.2	(f) -28.3	(g) -28.8	(h) 0.3	(i) -25.4	(j) 14.0	(k) -10.3	(l) 12.5	(m) -10.6	(n) 2.7	(o) 320	(p) 0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	31.4	22.1	29.9	21.1	27.9	20.1	24.6	28.9	23.3	27.8	22.1	26.3	6.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 23.0	(b) 18.4	(c) 26.9	(d) 22.2	(e) 17.5	(f) 26.2	(g) 20.9	(h) 16.1	(i) 25.2	(j) 76.2	(k) 29.0	(l) 70.9	(m) 27.9	(n) 66.1	(o) 25.8	(p) 30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.1	(b) 10.9	(c) 9.4	(e) -32.3	(f) 34.6	(g) 3.5	(h) 1.7	(i) -34.8	(j) 35.8	(k) -36.9	(l) 36.8	(m) -38.9	(n) 37.8	(o) -41.4	(p) 39.0		
(4) 12.1	(5) 10.9	(6) 9.4	(7) -32.8	(8) 26.4	(9) 3.5	(10) 2.1	(11) -35.3	(12) 27.9	(13) -37.4	(14) 29.1	(15) -39.4	(16) 30.2	(17) -42.0	(18) 31.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-11.8	-11.7	-2.2	5.9	13.6	18.9	21.4	19.7	15.6	7.6	-0.4	-9.1	
	DBStd	13.22	7.92	7.29	7.37	5.82	5.14	3.37	3.08	3.14	4.59	5.01	6.23	6.96	
	HDD10.0	2858	674	609	383	144	23	0	0	0	7	109	315	593	
	HDD18.3	4870	933	842	638	374	163	33	9	21	106	337	563	852	
	CDD10.0	1290	0	0	3	20	135	267	352	301	176	33	2	0	
	CDD18.3	259	0	0	0	0	17	50	103	63	24	2	0	0	
	CDH23.3	2545	0	0	1	7	223	464	987	568	261	34	0	0	
	CDH26.7	714	0	0	0	1	73	113	297	152	69	9	0	0	
Wind	WSAvg	4.3	4.5	4.5	4.7	5.0	4.9	4.1	3.4	3.4	4.1	4.7	4.6	4.4	
Precipitation	PrecAvg	561	15	10	24	54	74	96	85	68	60	46	18	12	
	PrecMax	922	62	31	69	192	201	223	212	217	186	144	45	51	
	PrecMin	211	1	0	5	3	10	18	11	8	7	2	0	0	
	PrecStd	128	14	7	13	37	47	53	48	40	44	39	12	10	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	7.9	19.0	23.8	32.2	32.4	33.8	32.5	32.1	27.6	17.6	7.2	
		MCWB	2.2	5.8	13.0	13.2	18.2	20.7	25.5	22.5	22.1	17.6	11.2	4.2	
	2%	DB	2.1	2.5	14.1	21.0	28.7	29.8	31.5	30.2	28.6	22.5	13.9	2.6	
		MCWB	1.0	1.0	10.3	11.9	17.5	19.8	24.0	22.4	19.7	14.2	9.1	0.9	
	5%	DB	0.3	1.0	10.9	17.7	25.9	27.6	29.9	27.9	26.9	18.0	11.0	1.1	
		MCWB	-0.2	0.1	8.0	10.6	15.4	18.5	22.4	21.1	18.7	12.3	7.6	0.3	
	10%	DB	-0.9	-0.8	7.2	15.2	22.7	26.2	27.8	26.5	23.9	15.9	7.6	0.0	
		MCWB	-1.2	-1.4	5.1	9.0	14.4	18.2	21.1	19.7	17.6	10.5	5.1	-0.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	6.9	14.1	15.7	22.6	24.6	28.0	25.5	24.3	19.6	12.8	4.5
			MCDB	4.0	8.7	17.0	21.4	27.4	29.0	32.4	28.8	29.7	25.2	16.4	6.9
2%		WB	0.8	1.3	10.5	13.2	19.0	22.5	25.5	23.7	22.0	15.3	9.7	1.3	
		MCDB	1.4	2.4	13.6	18.6	24.3	26.5	29.4	28.0	26.1	21.7	13.7	2.4	
5%		WB	-0.1	-0.1	7.4	11.5	17.4	20.8	23.9	22.5	19.9	13.0	7.5	0.2	
		MCDB	0.3	0.4	9.7	16.4	21.9	24.6	28.1	26.9	24.1	17.0	10.8	0.7	
10%		WB	-1.2	-1.4	4.8	9.9	15.8	19.4	22.5	21.2	18.3	11.2	5.2	-0.7	
		MCDB	-0.8	-0.8	7.3	14.7	21.6	24.2	26.9	25.0	22.8	15.0	7.6	-0.3	
Mean Daily Temperature Range		5% DB	MDBR	9.5	10.0	8.8	10.5	12.0	11.1	11.9	11.6	12.2	10.3	8.9	8.5
			MCDBR	9.5	10.5	13.2	15.9	16.5	13.6	13.4	13.4	15.3	15.1	13.3	9.7
	MCWBR		8.7	9.3	9.0	8.7	8.0	6.5	6.9	6.8	7.4	8.7	8.8	8.3	
	5% WB	MCDBR	9.0	10.1	12.3	13.9	12.7	11.1	12.4	11.9	12.7	13.2	12.7	8.7	
		MCWBR	8.3	9.3	8.7	8.6	7.4	6.7	7.4	7.0	7.7	8.2	8.7	7.7	
Clear-Sky Solar Irradiance	taub	0.297	0.296	0.309	0.361	0.381	0.390	0.389	0.391	0.359	0.318	0.286	0.267		
	taud	2.264	2.323	2.423	2.331	2.346	2.367	2.357	2.346	2.424	2.514	2.510	2.415		
	Ebn at Noon	787	879	922	896	886	876	872	855	856	849	820	802		
	Edh at Noon	80	93	98	118	121	119	119	115	98	77	63	62		
All-Sky Solar Radiation	RadAvg	1.70	2.82	3.85	4.90	5.51	6.16	6.48	5.43	4.12	2.58	1.64	1.22		
	RadStd	0.18	0.30	0.37	0.38	0.38	0.38	0.26	0.48	0.27	0.35	0.18	0.13		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page